

Introduction to Classification

hWhat is Classification in Machine Learning?

Classification is a supervised machine learning technique used to predict a category .

****Example:**** Predicting whether a loan application will be ****Approved**** or ****Rejected****.

How is it Different from Regression?

Classification	Regression
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Predicts categories or labels.	Predicts continuous numerical values.
Output is discrete (e.g., Yes/No).	Output is a number (e.g., price).
Example: Spam or Not Spam.	Example: House price prediction.

Real-Life Examples

- ****Classification:**** Predicting whether an email is ****Spam**** or ****Not Spam****.
- ****Regression:**** Predicting the ****price of a house**** based on its features.

Classification Algorithms

Logistic Regression

****Definition:****

Logistic Regression is a classification algorithm that predicts the probability that an input belongs to a particular class.

****How it Works****

- Calculates the probability of each class.
- Uses a sigmoid function to produce values between 0 and 1.
- Classifies based on a probability threshold (usually 0.5).

****Real-World Use Case****

- Loan approval prediction.

****Advantages****

- Simple and easy to interpret.
- Fast to train.
- Works well for binary classification.

****Limitations****

- Assumes a linear relationship.
- Cannot model complex patterns well.

Decision Trees

****Definition:****

A Decision Tree classifies data by asking a series of decision questions and splitting the data into branches.

****How it Works****

- Splits data based on the best feature.
- Creates branches until reaching a final decision (leaf node).

****Real-World Use Case****

- Customer churn prediction.

****Advantages****

- Easy to understand and visualize.
- Handles numerical and categorical data.
- Captures non-linear relationships.

****Limitations****

- Can overfit the training data.
- Sensitive to small changes in data.

Random Forest

****Definition:****

Random Forest is an ensemble learning algorithm that combines many decision trees to improve prediction accuracy.

****How it Works****

- Builds multiple decision trees using random subsets of data.
- Combines their predictions using majority voting.

****Real-World Use Case****

- Credit risk assessment.

****Advantages****

- High accuracy.
- Reduces overfitting.
- Works well with large datasets.

****Limitations****

- Slower than a single decision tree.
- Harder to interpret.

Comparison of Algorithms

Algorithm	Basic Idea	Use Case	Advantages	Limitations
Logistic Regression	Predicts probability using a sigmoid function	Loan approval	Simple, fast, interpretable	Assumes linear relationship
Decision Tree	Splits data into branches	Customer churn	Easy to understand	Can overfit
Random Forest	Combines many decision trees	Credit risk prediction	Accurate, reduces overfitting	Less interpretable, slower

Classification Metrics

Accuracy

****Definition:****

Accuracy measures the percentage of correct predictions out of all predictions.

****Formula****

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Accuracy = (Correct Predictions) / (Total Predictions)

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Precision

****Definition:****

Precision measures how many predicted positive cases are actually positive.

****Formula****

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$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

'''

Recall

****Definition:****

Recall measures how many actual positive cases are correctly identified.

****Formula****

'''

$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

'''

F1-Score

****Definition:****

F1-Score combines Precision and Recall into a single metric.

****Formula****

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$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

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Confusion Matrix

****Definition:****

A Confusion Matrix is a table that compares actual values with predicted values.

	Predicted Positive	Predicted Negative	
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	Actual Positive	True Positive (TP)	False Negative (FN)	
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	Actual Negative	False Positive (FP)	True Negative (TN)	
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Comparison of Metrics

	Metric	What it Measures	Sensitive to Class Imbalance	
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	Accuracy	Overall correctness	Yes	
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	Precision	Correctness of positive predictions	No	
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	Recall	Ability to find actual positives	No	
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| F1-Score | Balance between Precision and Recall | Less sensitive |
| Confusion Matrix | Counts TP, FP, TN, FN | Helps analyze imbalance |

Class Imbalance

Why Can Accuracy Be Misleading?

When one class appears much more often than another, a model can achieve high accuracy simply by always predicting the majority class while failing to identify the minority class.

****Example:****

If 95% of loan applications are approved, predicting ****Approved**** for every application gives 95% accuracy but completely misses rejected loans.

When to Prioritize Precision or Recall

Prioritize Precision

Use Precision when approving a risky loan would be costly.

****Loan Example:****

If a model predicts a loan should be approved, you want to be highly confident that the applicant is actually eligible.

Prioritize Recall

Use Recall when it is important to identify as many qualified applicants as possible.

****Loan Example:****

A high Recall ensures that most eligible applicants are approved, reducing the chance of rejecting good customers.

Real-World Case Study

Loan Approval Prediction

Goal

Predict whether a loan application should be approved based on applicant information.

Data Used

- Applicant income
- Credit score
- Employment history
- Loan amount
- Previous loan defaults
- Collateral information

Classifier Used

****Random Forest Classifier****

Key Results

- Achieved high prediction accuracy.

- Reduced overfitting compared to a single decision tree.
- Identified credit score, income, and previous defaults as the most important features for loan approval decisions.

Insights

Classification models help financial institutions make faster and more consistent loan approval decisions while reducing lending risk.